



**FACULTY OF ENGINEERING, BUILT ENVIRONMENT, AND
INFORMATION TECHNOLOGY (FOEBEIT)**

BACHELOR OF COMPUTER SCIENCE (BCS-ODL)

BCL1233 – SYSTEM ANALYSIS AND DESIGN

ASSIGNMENT 2

Process Modeling for a Hybrid Work Monitoring System

INSTRUCTIONS TO CANDIDATE:

1. This is an **INDIVIDUAL** assessment.
2. This assignment contributes **20%** of the total course mark.
3. Submit your completed assessment via **SPARK by 11:59 PM on 4 July 2026**.

WARNING

You are required to submit your work within the bounds of the University Infringement of Assessment Regulations. Plagiarism, paraphrasing and downloading large amounts of information from external sources will not be tolerated and will be dealt with severely.

SEGi University, Kota Damansara takes allegations of plagiarism very seriously. Submissions involving plagiarism will be marked but given zero mark. Plagiarism is the attempt to pass off the work of another as your own. Information taken from the work of others should be acknowledged by reference to obviate the charge of copying.

The work must be entirely your own. You will receive ZERO (0) for work that is not your own. The safety of your assessments is your responsibility. You must not permit another student access to your work.

LEARNING OUTCOMES

CLO2: Apply system analysis modelling techniques to represent system processes and interactions.

TASK DESCRIPTIONS

Based on the Hybrid Work Monitoring System developed in Assignment 1, continue your role as a system analyst and develop process models that illustrate how information flows through the system and how users interact with system functions.

REPORT REQUIREMENTS

Students must prepare a structured **System Modelling Report** based on the Hybrid Work Monitoring System. All diagrams must be properly labelled and supported with brief explanations.

1. Context Diagram (15 marks)

- a) Develop a Context Diagram showing the system boundary of the Hybrid Work Monitoring System.
- b) Identify and include all external entities interacting with the system.
- c) Show and label all data flows between the system and external entities.
- d) Provide a brief explanation of the system boundary and external entities (100–150 words)

2. Data Flow Diagram (DFD) Level 0 (25 marks)

- a) Develop a Level 0 DFD showing the main processes of the system which include:
 - Processes
 - Data flows
 - Data stores
 - External entities
- b) Briefly describe each main process.
- c) List all the data stores and explain their purpose.

3. Data Flow Diagram (DFD) Level 1 (25 marks)

- a) Select ONE (1) main process from the Level 0 DFD.
- b) Decompose it into a Level 1 DFD showing detailed subprocesses.
- c) Ensure logical flow and proper decomposition of the selected process.

4. Use Case Diagram (20 marks)

- a) Identify all relevant actors in the system.
- b) Develop a Use Case Diagram showing system functionalities.
- c) Include appropriate relationships such as include, extend, or generalization where applicable.

5. Use Case Description (15 marks)

- a) Select ONE (1) use case from the Use Case Diagram.
- b) Develop a detailed Use Case Description.

EVALUATION CRITERIA

Assessment Component	Description	Mark Allocation
Context Diagram	Correct identification of system boundary, external entities, and data flows. Diagram is clear, consistent, and properly labelled.	15%
DFD Level 0	Accuracy of system processes, data flows, data stores, and external entities. Logical structure and consistency with Context Diagram.	25%
DFD Level 1	Correct decomposition of selected process with clear subprocesses and logical data flow representation.	25%
Use Case Diagram	Correct identification of actors and system functionalities with proper relationships (include/extend/generalization).	20%
Use Case Description	Completeness and clarity of selected use case including all required elements (preconditions, flow, postconditions).	15%
TOTAL		100%

SUBMISSION REQUIREMENTS

1. A complete report in **Word or PDF format**.
2. Structured answers covering all tasks and questions.
3. All diagrams must be neat, labelled, and readable.
4. Use appropriate modelling tools (e.g., Draw.io, Visio, Lucidchart, or handwritten scanned clearly)
5. Provide brief explanations for each section.
6. Submission must be uploaded via **SPARK** on or before the given deadline.